

Enclave Compliance Suite

Author: Executive & Lead Architect	Target Standards: NIST SP 800-171 R2/R3, CMMC 2.0 Level 2, NIST AI RMF
Products Covered: Enclave-SSP, Enclave-GAP, Enclave-AI	Deployment Boundary: 100% Air-Gapped / On-Premise Desktop Engines

1. Executive Summary & Core Product Portfolio

Enclave Compliance Solutions is designed as a suite of privacy-first, on-premise desktop applications engineered for defense contractors and regulated enterprises. Cloud AI models introduce unacceptable regulatory risks under **DFARS 252.204-7012** due to potential leaks of Controlled Unclassified Information (CUI). The Enclave suite eliminates third-party cloud data egress by embedding local LLM orchestration engines directly into the client's desktop environment.

- **Enclave-SSP:** Automated System Security Plan (SSP) implementation statement writer for NIST SP 800-171 / CMMC Level 2 audits.
- **Enclave-GAP:** Automated policy & procedure gap analyzer utilizing a local Ollama sidecar to evaluate documentation against all 320+ NIST SP 800-171A assessment objectives.
- **Enclave-AI:** Governance and risk mapping platform engineered for the NIST AI Risk Management Framework (AI RMF).

2. Air-Gapped Local LLM Strategy (Ollama Engine)

By leveraging open-weight models (Qwen 2.5, Llama 3.1) via a local Ollama instance, Enclave products maintain an absolute zero-trust privacy boundary. CUI and internal IT policy data never leave the customer's hardware.

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[ Internal System Policies & Infrastructure Docs ] + ==> [ RAG Vector Database ] [ NIST SP 800-171 & CMMC Assessment Guides ] || √ [ Ollama Local Model (Llama 3.1 / Qwen 2.5) ] || √ [ Compliant Implementation Statement ]
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Recommended Hardware & Base Models

- **Standard Workstations (8GB - 16GB VRAM / RAM):** qwen2.5:14b or llama3.1:8b
- **High-Performance Enclaves (24GB+ VRAM):** qwen2.5:32b or llama3.1:70b-instruct-q4_K_M

Custom Modelfile for CMMC SSP Control Writing (`Modelfile.cmmc`)

```
FROM qwen2.5:14b

PARAMETER temperature 0.2
PARAMETER top_p 0.9

SYSTEM ""
You are an expert Certified CMMC Assessor (CCA) and NIST SP 800-171 Compliance Architect.
Your goal is to draft precise, audit-ready Implementation Statements for System Security Plans (SSPs).

For every control provided, format your response using this exact structure:
1. Control ID & Name: State the control family and ID.
2. Control Requirement Summary: Briefly summarize what NIST SP 800-171 mandates.
3. Implementation Statement: Detail HOW the control is satisfied in practice (mentioning roles, mechanisms, tools, and configurations). Use direct, declarative active voice.
4. Assessment Objective Checklist: Address each assessment objective from NIST SP 800-171A.
```

5. Evidence Artifacts: List 3-5 specific artifacts an auditor will ask for (e.g., policy docs, screenshots, log outputs).

Rule: Never make assumptions about tools or architecture not provided in the prompt context. If details are missing, explicitly highlight what needs input in brackets like [INSERT TOOL/ROLE HERE].
""

3. Automated Policy Gap Analysis (NIST SP 800-171A)

CMMC assessors do not evaluate the high-level 110 requirements directly; they assess against the **320+ sub-objectives** in **NIST SP 800-171A**. Enclave-GAP evaluates policies at this granular level.

NIST SP 800-171A Objective Breakdown Example (Control 3.1.1)

- **[a]** Authorized users are identified.
- **[b]** Processes acting on behalf of authorized users are identified.
- **[c]** Devices acting on behalf of authorized users are identified.
- **[d]** System access is limited to authorized users.
- **[e]** System access is limited to processes acting on behalf of authorized users.
- **[f]** System access is limited to devices acting on behalf of authorized users.

Modelfile for Gap Analysis (`Modelfile.gap-analysis`)

```
FROM qwen2.5:14b
```

```
PARAMETER temperature 0.1
```

```
PARAMETER top_p 0.8
```

```
SYSTEM ""
```

```
You are a Lead CMMC C3PAO Assessor conducting a Policy & Procedure Gap Analysis against NIST SP 800-171A.
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```
Your task is to compare the provided POLICY DOCUMENT snippet against the provided NIST SP 800-171A CONTROL OBJECTIVES.
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```
For each assessment objective, evaluate the policy and output a markdown table with this strict structure:
```

1. Objective ID: The sub-objective letter (e.g., [a], [b]).
2. Status: [MET], [PARTIALLY MET], or [NOT MET].
3. Policy Finding: Quote or reference the exact section of the policy that satisfies the objective.
4. Gap Analysis & Remediation: If NOT MET or PARTIALLY MET, explain precisely what language or clause is missing and provide a draft sentence to add to the policy to satisfy the gap.

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Rules: Be strict. Implicit references do NOT count as MET; policies must explicitly state requirements.
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4. Enclave-GAP Desktop Software Architecture

Enclave-GAP executes as a native desktop application (Tauri or Electron) that orchestrates a lightweight, embedded Ollama engine as an isolated sidecar process bound strictly to `127.0.0.1:11434`.



5. Umbrella Brand & Commercialization Strategy

To establish market credibility, all three tools are housed under **Enclave Compliance Solutions Inc.** with domain routing pointing back to a central hub.

Product	Target Market	Primary Value Proposition
Enclave-SSP	Defense Industrial Base (DIB) Contractors	Generate C3PAO-ready SSP documentation without cloud CUI exposure.
Enclave-GAP	CISO / Compliance Consultants	Instant 800-171A policy objective mapping in seconds via local AI.
Enclave-AI	Enterprise AI Governance Teams	Map, measure, and manage NIST AI RMF compliance locally.

6. Legal Disclaimers & Licensing Compliance

Required EULA & Marketing Terms

Non-C3PAO Certification Disclaimer: Enclave Compliance Solutions provides software tools to assist in documentation and gap identification. Use of Enclave-SSP, Enclave-GAP, or Enclave-AI does not guarantee a successful audit or formal certification by a C3PAO or NIST assessor.

- **Open Source Licenses:** Ollama (MIT), Qwen 2.5 (Apache 2.0), Llama 3.1 (Llama 3 Community License with *"Built with Llama"* attribution).
- **UCC Warranty Disclaimer:** Software provided "AS IS" without implied warranties of merchantability or fitness for a particular purpose.
- **Limitation of Liability:** Liability capped at fees paid over the preceding 12 months; indirect/consequential damages strictly waived.
- **Human-in-the-Loop Requirement:** End-user agreements mandate that all LLM outputs undergo mandatory human review by a qualified Security Officer.